UI, Ideation & Prototypes

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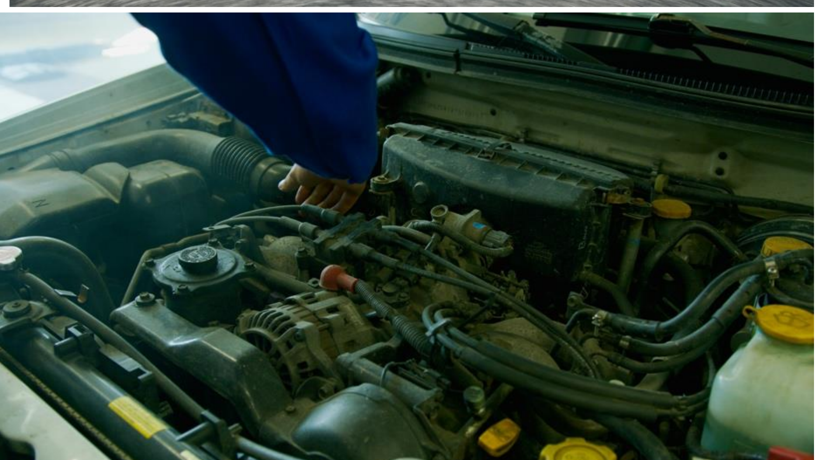
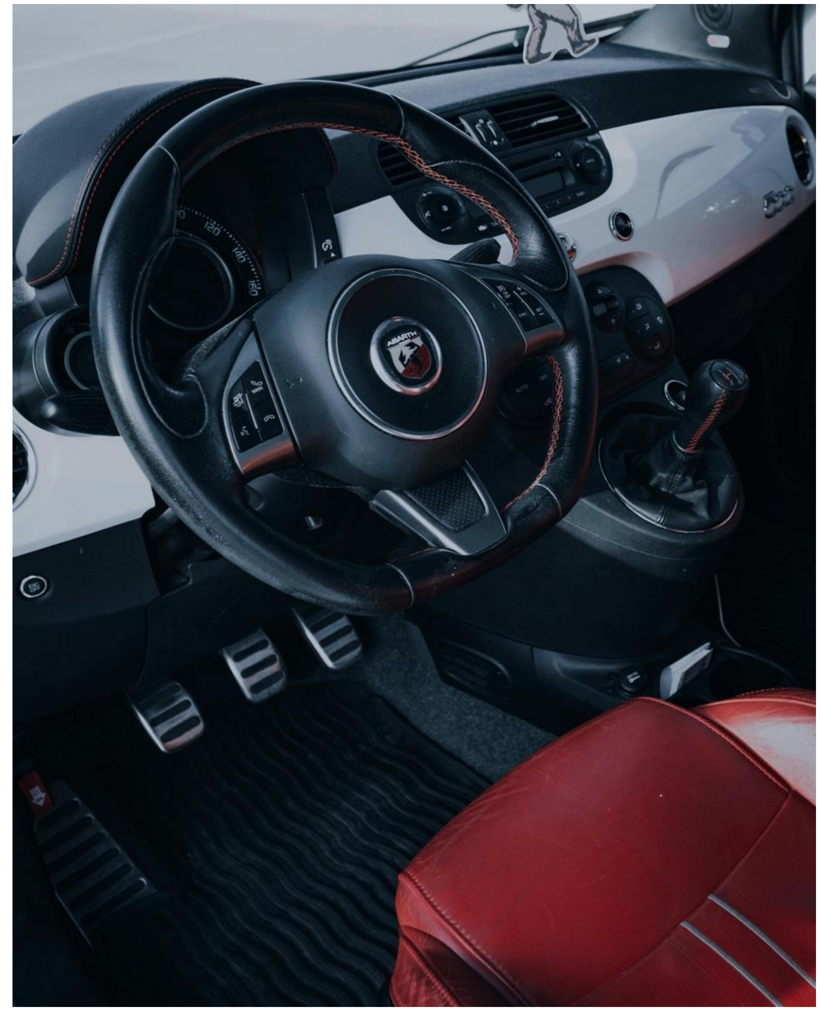
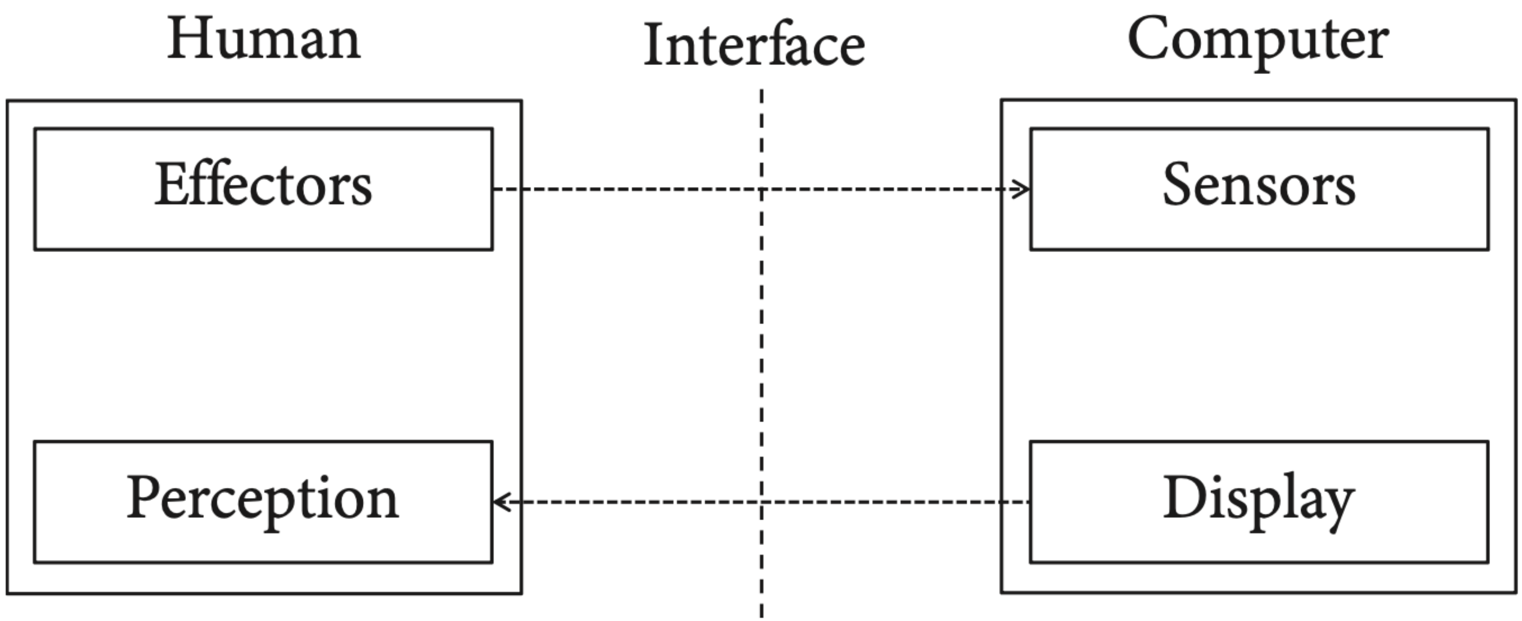
The user perspective



VS



Human-Computer System

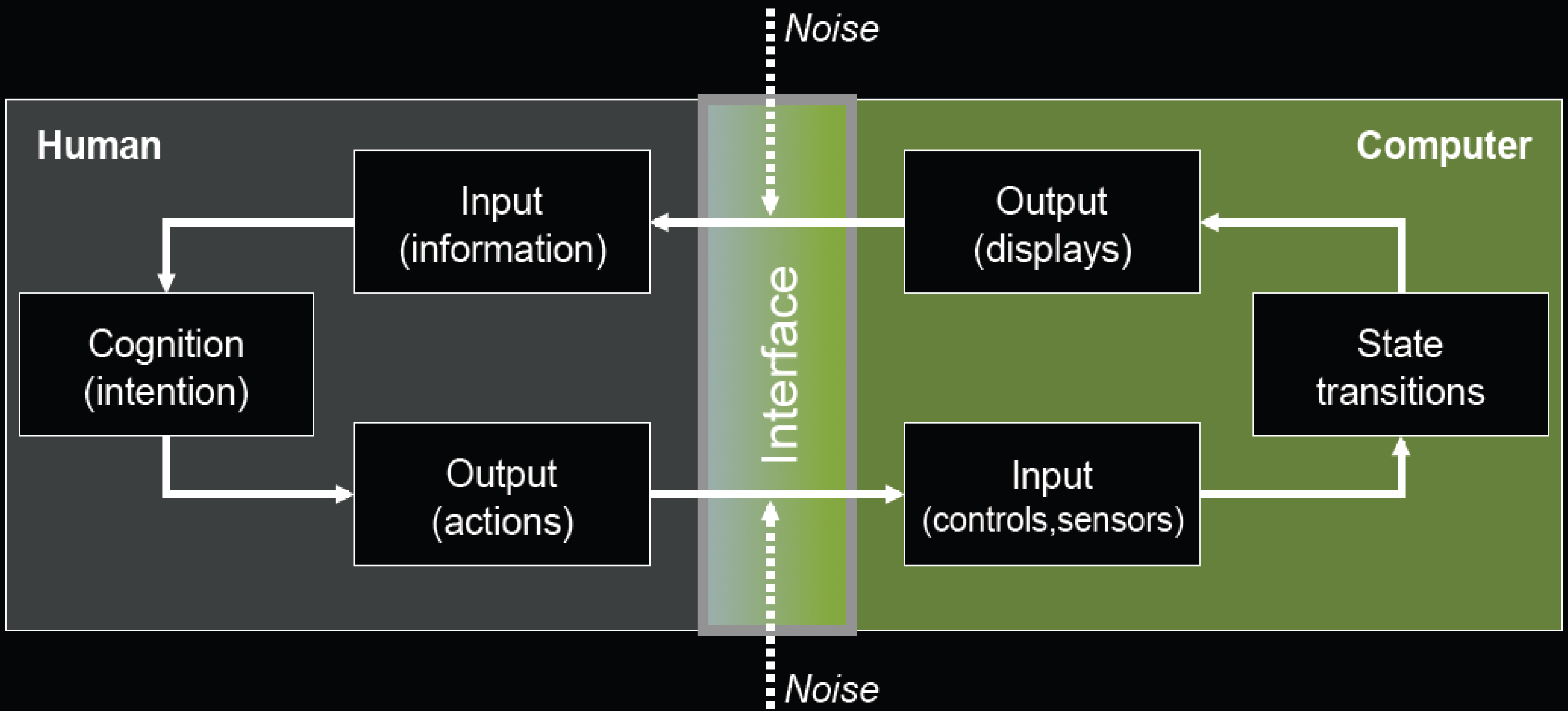


User Interface (UI)

User Interface: users can provide input and instructions to a computer and receive feedback from it. It enables interaction with a computer.

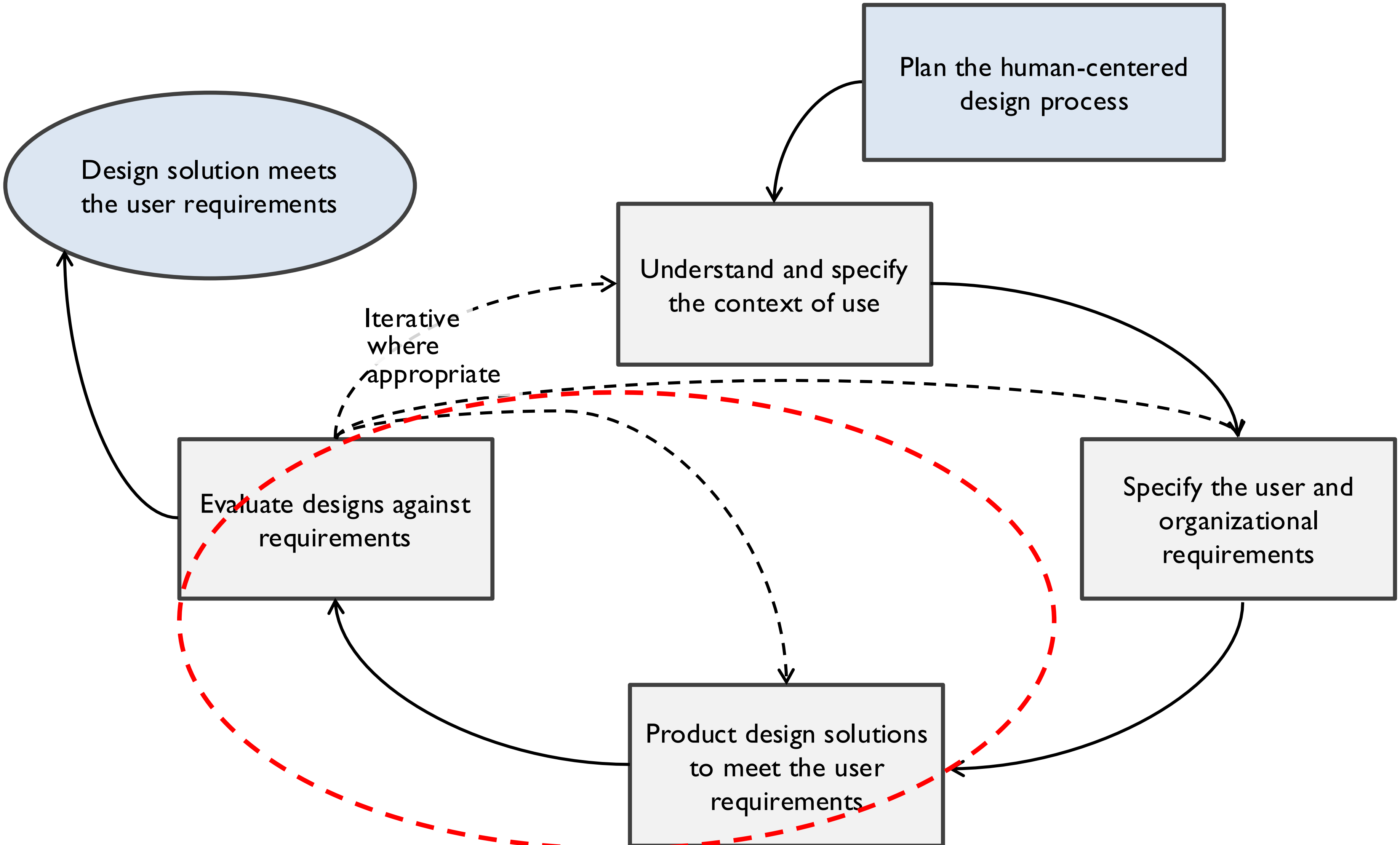
Simplifies the underlying technical system for the user: It allows users to command a computer without bothering with what it does under the hood.

The interaction loop



Hollnagel & Woods 2005, Sanders 1998, Seow 2005, Wickens & Hollands 2007, Wickens 1984

Human-Centered Approach



Creative Ideation

Producing alternative ideas that are of high quality.

Methods:

- Brainstorming
- Sketching
- Storyboards

Brainstorming

- Postpone criticism: By withholding criticism, the team can avoid throttling the generative process.
- Divergence: Encourage wild, diverse ideas.
- Quantity: Try to produce as many ideas as possible.
- Accumulation of knowledge: It is acceptable and encouraged to build on proposed ideas and known solutions.
- Equal significance: Every participant and every idea is equally valuable.

Design - Sketches

Hand sketching is the most natural way to express your ideas so that both you and collaborators can (be):

Evocative

Suggest

Explore

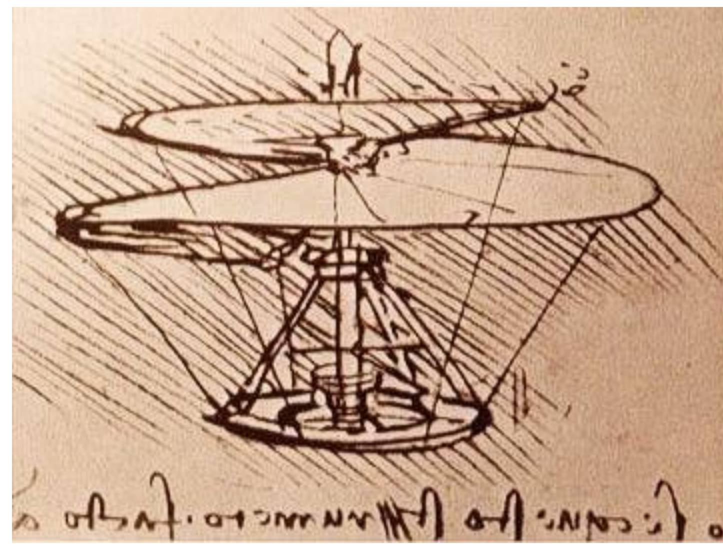
Question

Propose

Provoke

Tentative

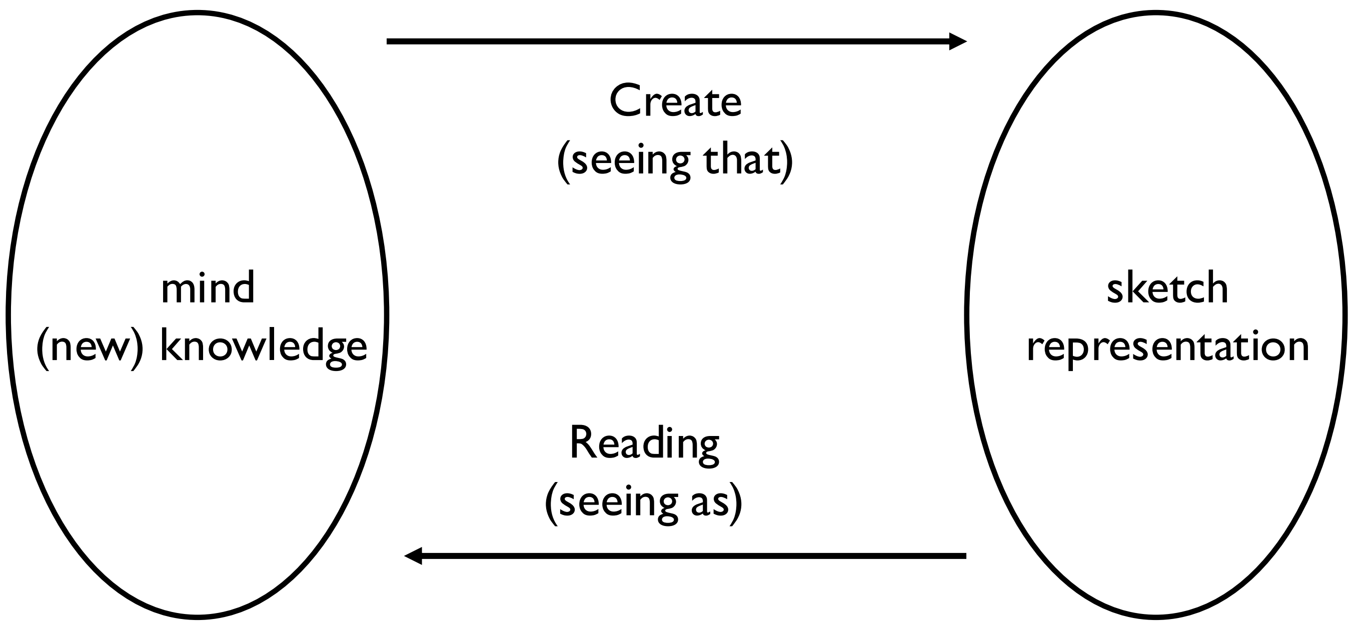
Noncommittal



Leonardo da Vinci: Helicopter

Sketching

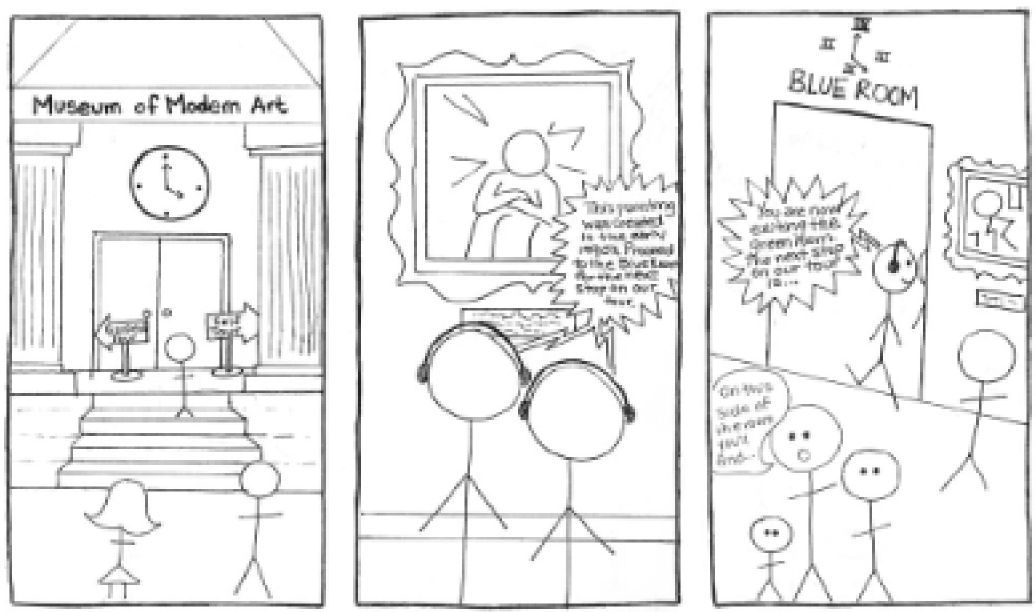
“as relating far more to an activity or process, rather than a physical object or artifact.”



Buxton, B. Sketching the User Experience

Design - Storyboards

Storyboarding is a technique (HCI context) for demonstrating system interfaces and **contexts** of use.



Storyboard of tour guide

http://dl.acm.org/citation.cfm?id=1142410

Prototype

The word *prototype* derives from the Greek πρωτότυπον *prototypon*, "primitive form", from πρῶτος *protos*, "first" and τύπος *typos*, "impression".

To produce an instance of an idea that serves as a model that can be assessed. The term proto implies a sense of incompleteness: When a prototype is created, it is accepted that it is not final. It may be missing functionality or entire components compared to a complete product.

Prototype

Simply to answer:

“What works and what doesn’t?”

Generally support creativity, communication and early evaluation.

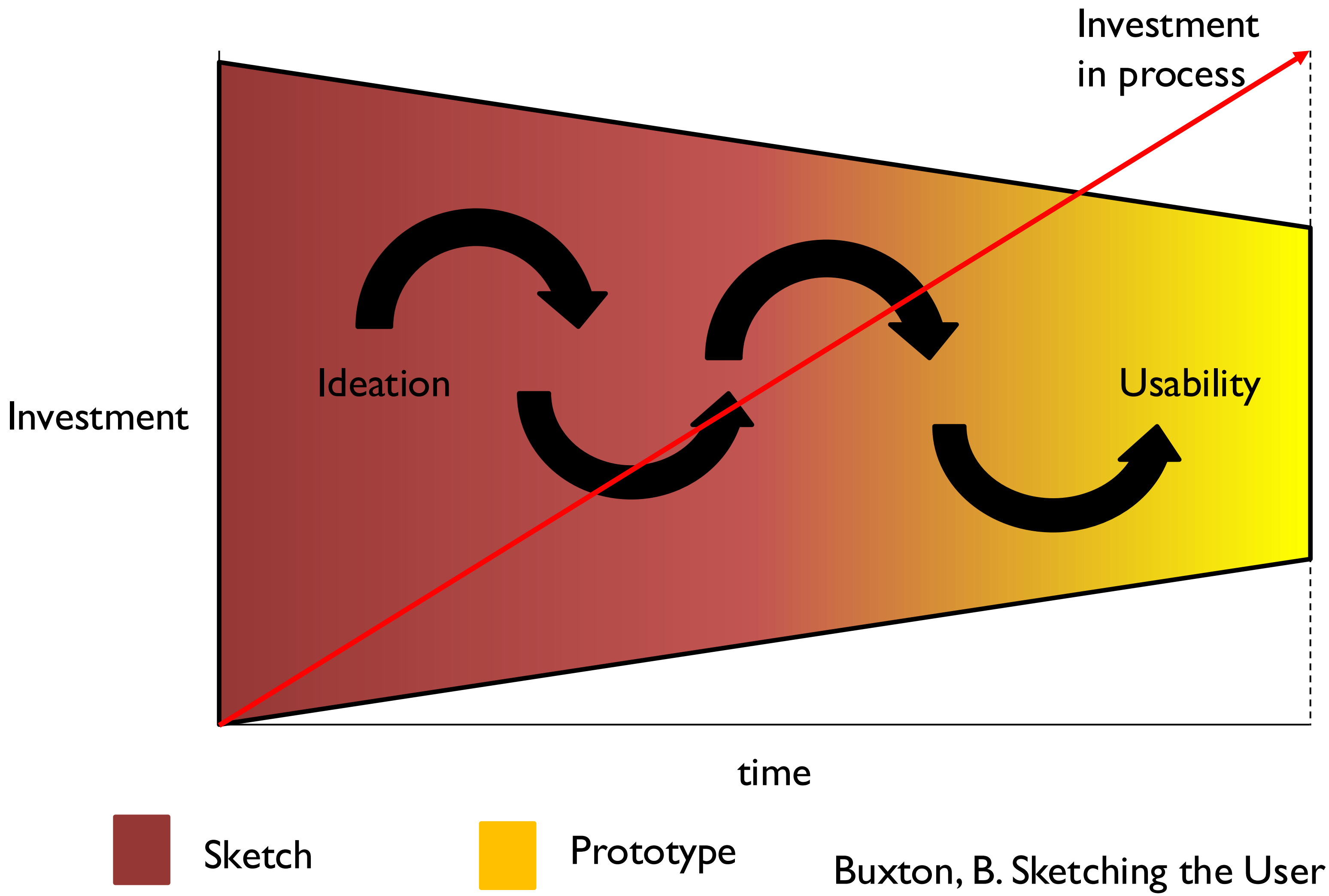
Prototype

Prototypes are used in HCI for two main purposes:

Studying the feasibility of an idea by creating it. Creating a prototype requires making key decisions, something that may still be avoided while ideating or sketching.

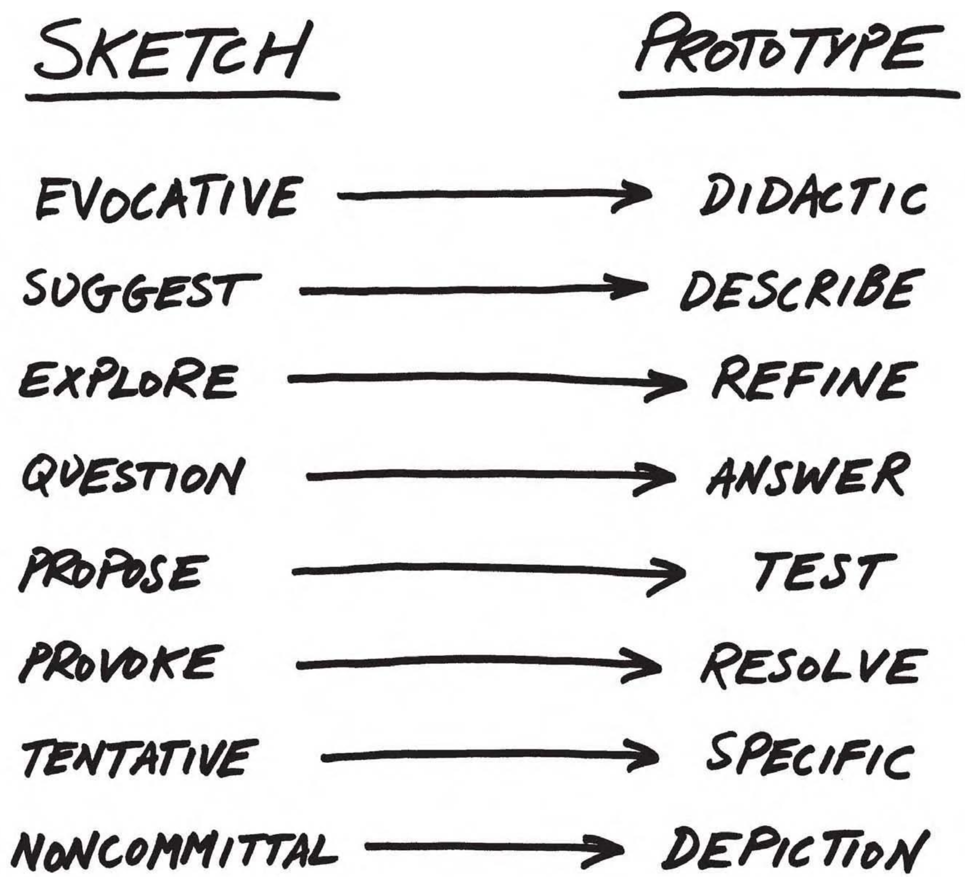
Presenting an idea in a concrete form to others, such as users and colleagues, to experience and test it.

Sketch vs Prototype



Buxton, B. Sketching the User Experience

Sketch vs Prototype



Buxton, B. Sketching the User Experience

Prototype – Low vs High Fidelity

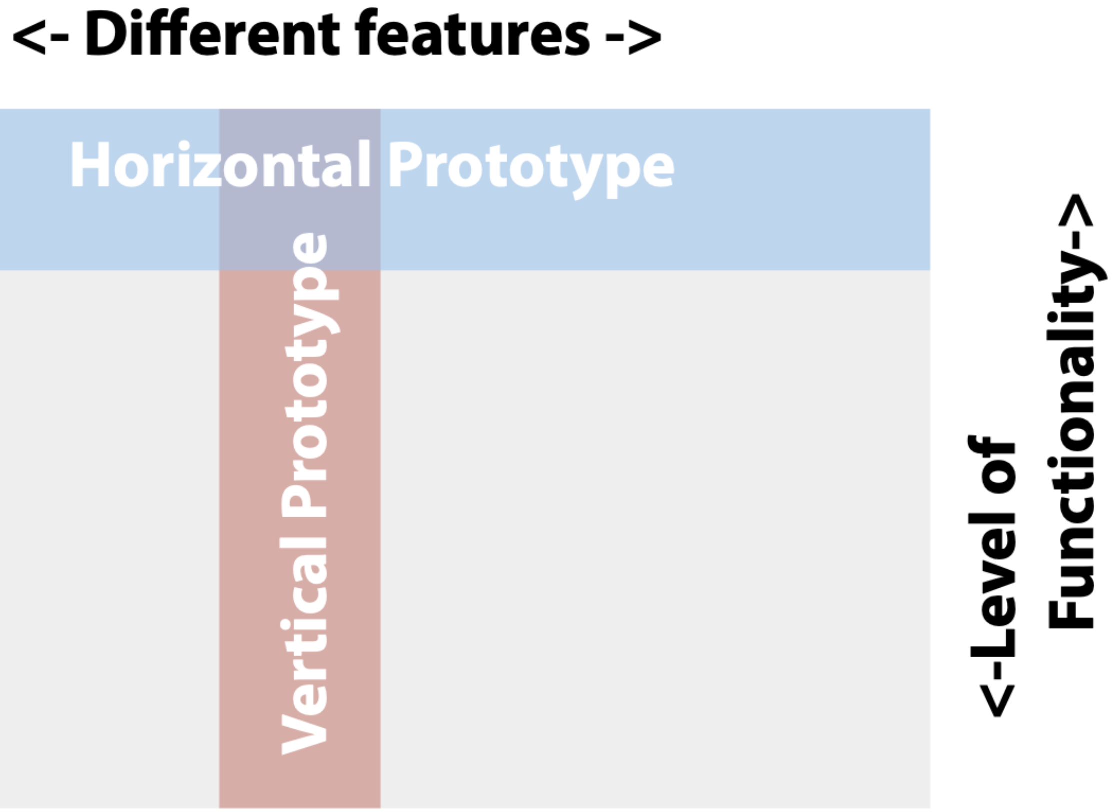
Low Fidelity (Lo-Fi): constructed quickly and provide limited or no functionality. (e.g., paper prototype)

High Fidelity (Hi-Fi): fully interactive (e.g., interactive mock-ups, software)

Prototype – Low vs High Fidelity

	Advantages	Disadvantages
Low-Fidelity (Lo-Fi)	Lower development cost. Evaluate multiple design concepts. Useful communication device. Address screen layout issues. Useful for identifying market requirements. Proof-of-concept.	Limited error checking. Poor detailed specification to code to. Facilitator-driven. Limited utility after requirements established. Limited usefulness for usability tests. Navigational and flow limitations.
High-Fidelity (Hi-Fi)	Complete functionality. Fully interactive. User-driven. Clearly defines navigational scheme. Use for exploration and test. Look and feel of final product. Serves as a living specification. Marketing and sales tool.	More expensive to develop. Time-consuming to create. Inefficient for proof-of-concept designs. Not effective for requirements gathering.

Vertical and Horizontal Prototypes



Based on: http://www.nngroup.com/articles/guerrilla-hci/

Horizontal Prototypes

Horizontal prototypes develop one entire layer of the design at the same time.

They address such issues like:

- consistency
- coverage
- redundancy

Vertical Prototypes

Vertical prototypes ensure that the designer can implement the full, working system, from the user interface layer down to the underlying system layer.

For example, a vertical prototype of a spelling checker for a text editor does not require text editing functions to be implemented.

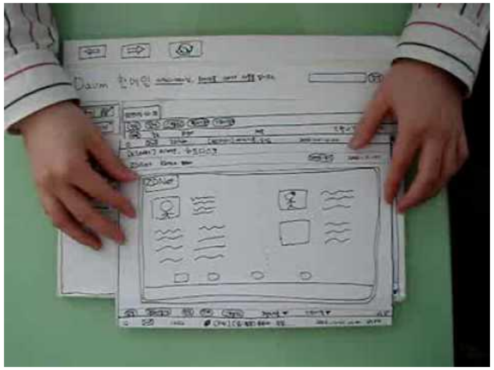
Functional Prototypes

A **functional prototype** captures both function and appearance of the intended design, though it may be created with different techniques and even a different scale from the final design.

Paper Prototype

Paper prototyping is a technique that allows you to create and test user interfaces quickly and cheaply. It's easier to change a prototype than the final design. (Lo-Fi Prototyping)

(https://www.nngroup.com/reports/paper-prototyping-training-video/)



Testing Session:

1 User

1 Facilitator

Rest of team as Observers

Marc Rettig. 1994. Prototyping for tiny fingers. *Commun. ACM* 37, 4 (April 1994), 21-27.

https://youtu.be/GrV2SZuRPv0

Paper Prototype - Interaction

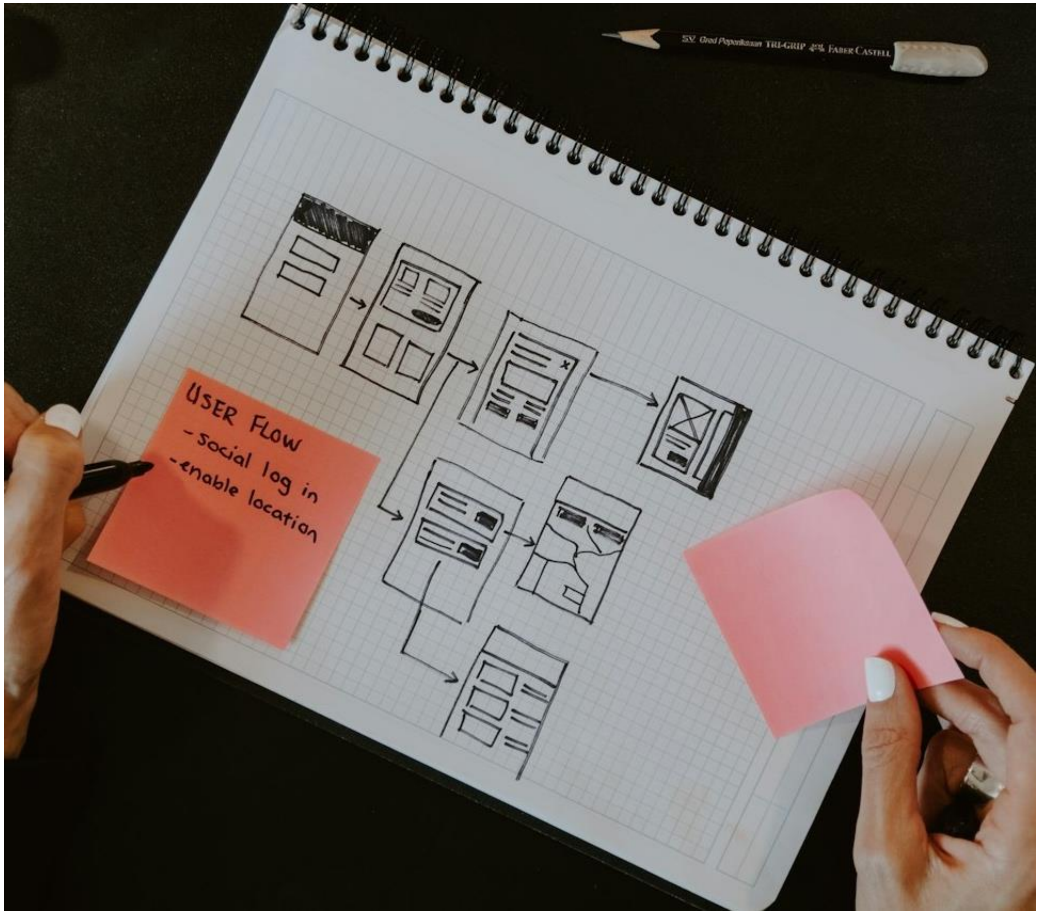
2D paper can be a strong limitation…. Use your imagination!
(But keep in mind: quick and cheap!)

Bifocal Display Example
https://youtu.be/DaF5brrdpJw

Cardboard Prototyping
https://www.youtube.com/watch?v=k_9Q-KDSb9o

Lo-Fi Prototyping - Wireframes

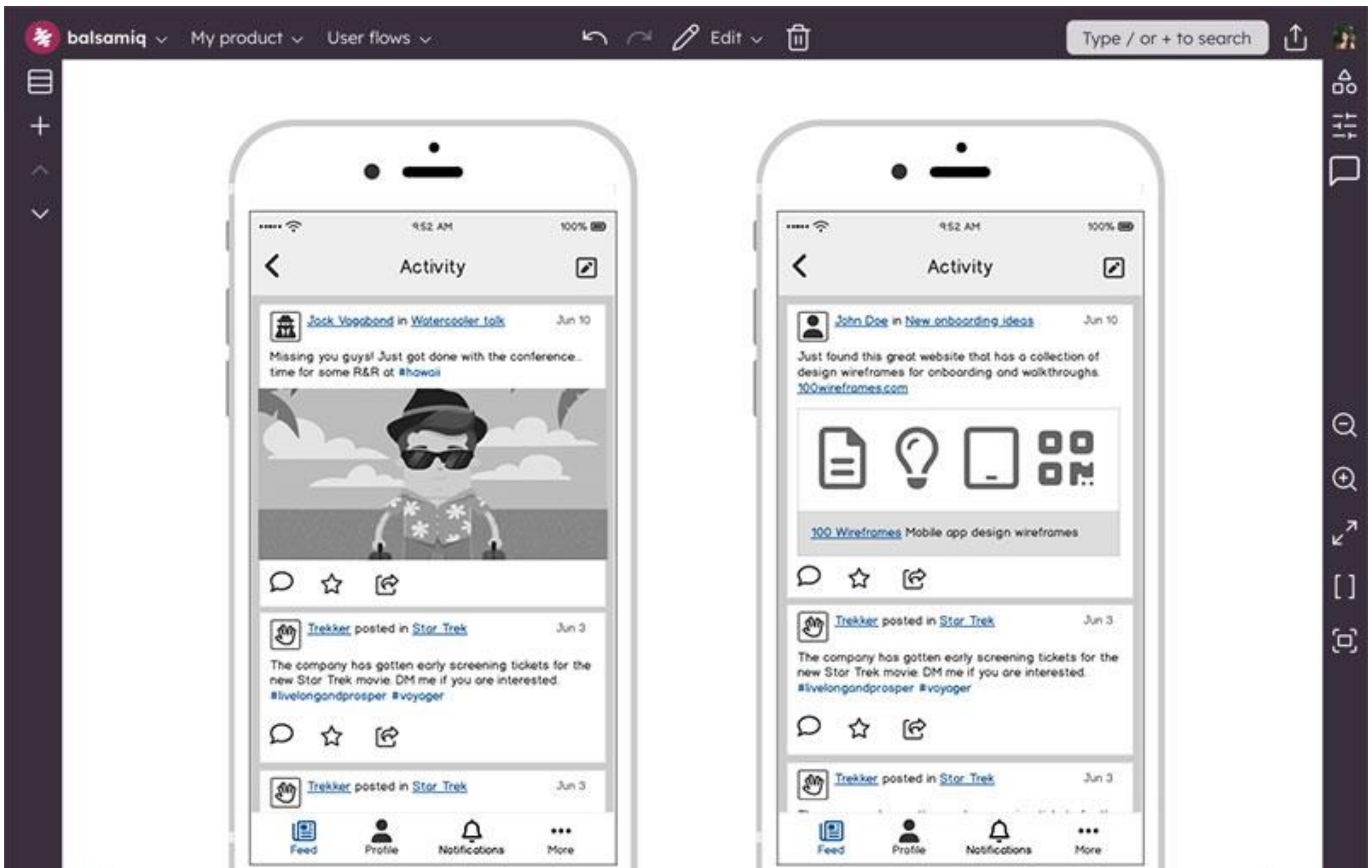
Basic visual representations of a user interface that outline the structure and layout of a webpage or app.



Wireframes (not so Lo-Fi version)

https://balsamiq.com/

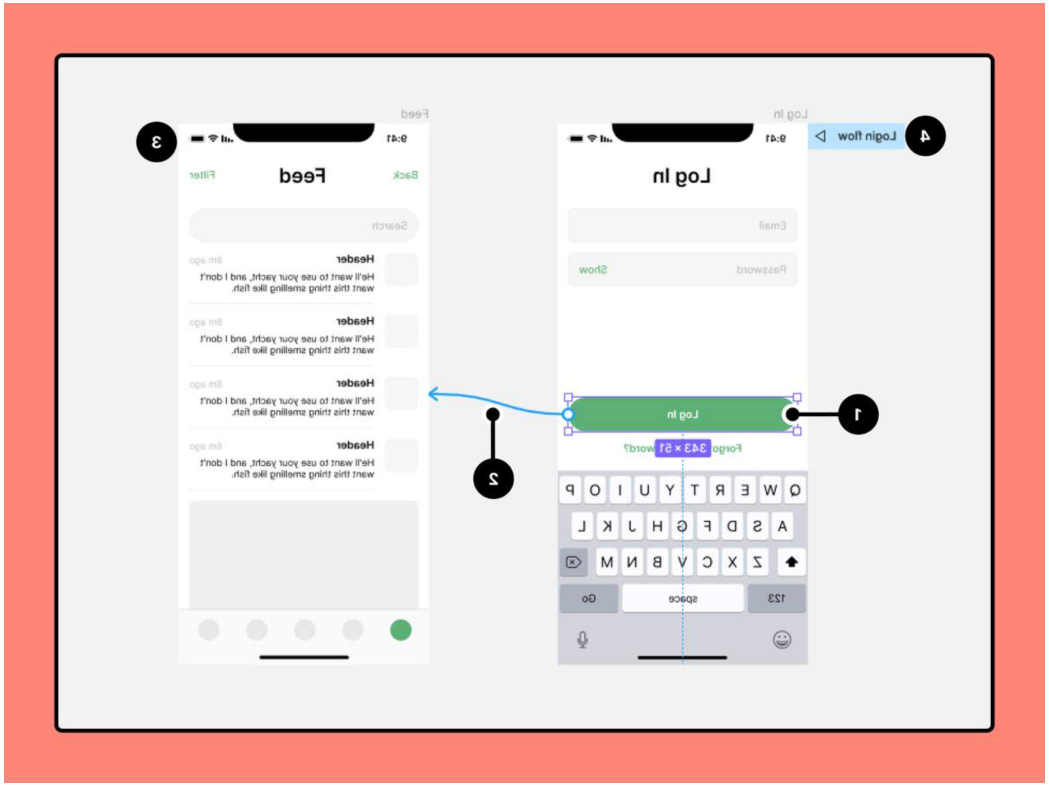
Rule of thumb: if you need to learn a tool that is not so Lo-Fi



Hi-Fi Prototyping - UI & Design

https://www.sketch.com/

https://www.figma.com/prototyping/



Usability

Usability concerns how easily computer-based tools may be operated by users trying to accomplish a task.

Usability differs from utility. Usability concerns whether users can use the product in a way that makes it possible to realize its utility; utility is about whether the goal is important to the user.

Usability

Is a **quality attribute** to assess interfaces considering:

Effectiveness: How good a product is at doing what it is supposed to do?

Efficiency: Once users have learned the design, how quickly can they perform tasks?

Safety: How many errors do users make, how severe are these errors, and how easily can they recover from the errors?

Utility: Does the product provide an appropriate set of functions that will enable users to carry out all their tasks in the way they want to do them?

Learnability: How easy is it for users to accomplish basic tasks the first time they encounter the design?

Memorability: When users return to the design after a period of not using it, how easily can they reestablish proficiency?

Usability Test

An evaluation of the usability of an interactive system with representative users doing representative tasks.

Users as evaluators

Tasks

The tasks, especially the first few tasks, should be easy.

Should be central to the users' real or imagined activities.

Should be expressed in terms of users' real or imagined activities.

It should be clearly stated when the task is completed.

Tasks can be open-ended so that they are in part specified by the user.

Think-Aloud Studies

Imagine you are to evaluate a computer system and put a prospective user in front of the system with a task they want to accomplish — and then nothing observable happens.

As they finally begin to use the system, they swiftly complete their task in silence.

You have learned (almost) nothing about what works or does not work in the user interface.

Think-Aloud Studies

The participants then verbalize their thinking when they interact with the system under evaluation or immediately afterward.

The verbalization and other notable events are captured and then analysed to derive insights about the thinking of the participants.

Think-Aloud Studies

Instructions:

Participants should "think aloud as if you were alone in this room."

(1) explain that the participant is not the object of the test—the system is,

(2) ensure that the participant is the expert and primary speaker,

(3) acknowledge thinking aloud through utterances such as "mm hmm," "yeah," and "ok."

Prompts: "keep talking,"; "What are you trying to achieve?" or "What are you thinking?"

Analysis

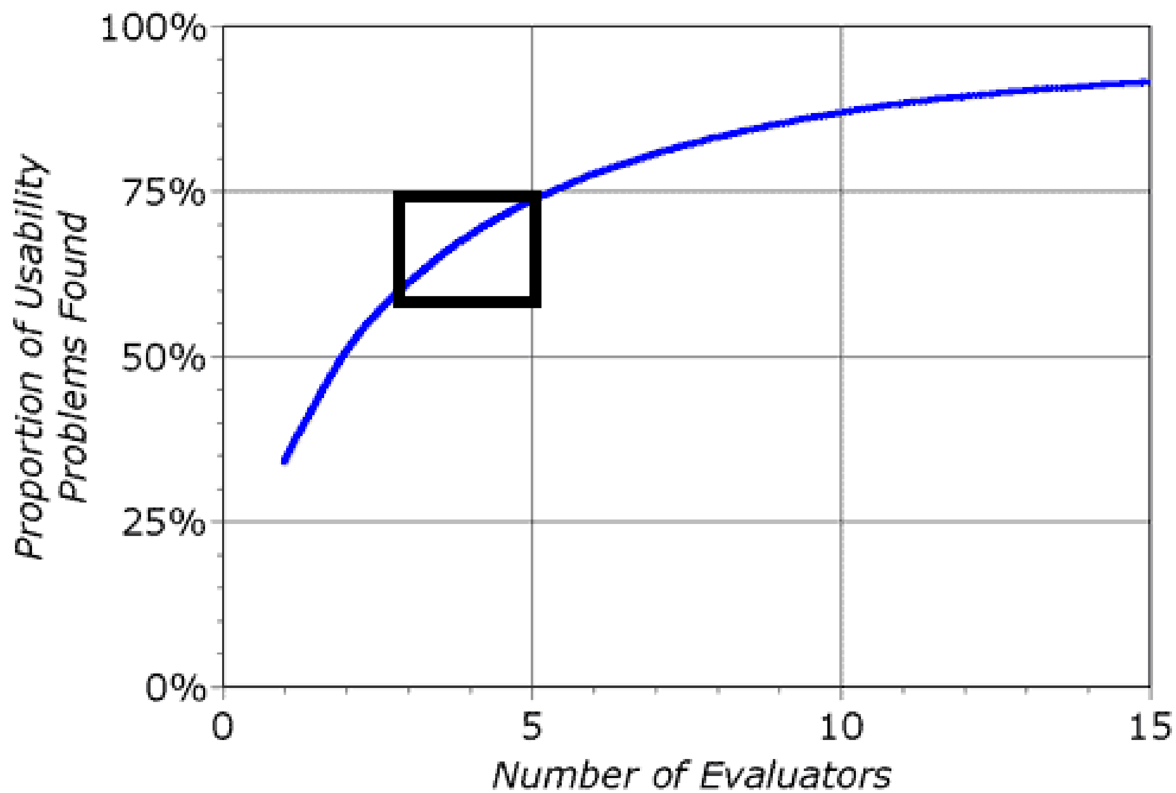
Failure to reach the goal: Examples: The user articulates a goal and cannot succeed in attaining it within three minutes; the user gives up; the user produces a result different from that expected or the task given; the system crashes or reaches a state from which the user cannot recover.

Misunderstandings about the system: Examples: The user knows what the goal is but cannot pick the right action; the user expresses surprise; the user expresses ways to improve the system.

Negative experience: Examples: The user expresses a negative feeling; the user says something is a problem; the user expresses a negative sentiment toward the system.

Basic Evaluation – Evaluators & Users

3-5 evaluators/users find 66-75% of usability problems



Basic Evaluation – Evaluators & Users

Novice

Intermediate

Experts

Basic Evaluation – Evaluators & Users

Number of evaluators/users needed for reliable usability issues detection:

Novice: 7 or +

Intermediate: 4 or +

Experts: 2

Expert (users or evaluators) evaluation is highly valuable.

Brainstorming

Writing down every idea that comes to mind, no matter what it is.

Exercise:

- 5 mins to produce sketches of your app (individual)
- 5 mins to discuss it with peers
- 5 mins to produce a storyboard about using your app (individual)
- 5 mins to discuss it with peers